FRAN SÁNCHEZ RODRIGO

Tools, Engine & Optimization Engineer

fran.sanchez.rodrigo@gmail.com LinkedIn profile Spain https://urkaz.github.io/

EXPERIENCE

Unreal Console Porting Specialist at **Making Port** *May 2026 - Present*

- Game porting using Unreal Engine 5 to console platforms.
    - The Relic: First Guardian (Switch 2, XSX)

Unreal Tools & Engine Programmer *Dec 2021 - Mar 2025*
Unreal Console Porting Specialist at **Catness Game Studios**

- Game porting using Unreal Engine 4 and 5 to console platforms:
    - Xuan Yuan Sword 7 (Switch)
    - Metamorphosis VR (Oculus Quest 2)
    - American Arcadia (Switch, PS4, XSX)
    - Click to see the full list
- Plugin and tool development to extend engine functionality and streamline console porting:
    - Project analysis and statistics tools.
    - Custom Blueprint nodes.
    - Context menu extension framework to create custom editor tasks.
    - Xbox user management and controller handling.
    - Optimization tools (automatic MaxDrawDistance, tick management, mesh instancing).
    - Automatic platform-based asset replacement.
- General engine modifications and fixes:
    - Indexed Actor tags optimization.
    - Mobile Software Occlusion fixes.
    - Improved MinLOD behavior for consoles.

Quality Assurance Technician at **Lollipop Robot** *Feb 2016 - Sep 2020*

- Playtesting and Compliance testing for PC and consoles (Xbox One, PS4, Switch).
- Experience using popular bug trackers like Jira, Mantis, Bitbucket, or GitHub.

EDUCATION

Master's Degree in Video Game Programming *2020 - 2021*
U-Tad (Madrid)

Graduate (Bachelor's Degree) in Video game Design and Development *2012 - 2016*
Universitat Jaume I (Castellón)

SKILLS AND COMPETENCIES

- Advanced proficiency in game programming with Unreal Engine (C++).
- Experience with version control systems (Git, Perforce).
- Fast adaptability to new codebases.
- Ability to work with low-level SDKs and systems.
- Experience with multiple programming and scripting languages: Java, C#, Python, Lua.
- Basic knowledge of CMake for C++ development.
- Intermediate level of game development with Unity.

LANGUAGES

- English: Professional working proficiency.
- Spanish: Native or bilingual proficiency.
- Catalan: Native or bilingual proficiency.

OTHER PROJECTS

Unused Node Cleaner

"Unused Node Cleaner" is an **Unreal Engine plugin** that allows you to clean Blueprints by removing unused nodes left behind in your graphs, reducing visual noise and improving readability.

- Fab.com download page

Orion Engine

Orion Engine is a C++20 game engine built from scratch, using The Cherno's Hazel series as a starting point to understand how a real engine is structured.

- Click for more details

Quantum fields

Entry for the 2023 Epic MegaJam. A portal-like game where you can open portals to see the past. I implemented an asynchronous dynamic mesh cutting system.

- Click for more details

ACADEMIC PROJECTS

Ronin-Hood Sisters — Master's Degree in Video Game Programming

Tactical RPG made with Unreal Engine 4 (C++), inspired by magical girls with elemental powers and destructive abilities.
I worked on programming the game world, a custom map creation tool, the fog of war system, the save system, and the logic behind multiple UI elements, including the costume selection screen, the inventory, and the combat HUD.

- Click for more details

Medieval Floor — Master's Degree in Video Game Programming

Action game made with Unreal Engine 4 (C++). Local cooperative game for up to 4 players, where you clear waves of increasing difficulty.
I worked on programming the gameplay flow, including enemy pool generation, wave-based spawning, and saving high scores to disk.

- Click for more details

Unity Zombie Killer — Master's Degree in Video Game Programming

First-person zombie survival game where players must complete three different quests to win. Developed with Unity (C#) in 5 days.
I worked on programming the weapon systems, pickable items, zombie AI, and the quest system.

- Click for more details